

# Entropy-Linked Autonomous Propagation and Self-Erasure (ELAPSE): A Multi-Domain Ensemble Framework for Active Data Containment

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**Abstract**—When sensitive data enters a peer-to-peer network, it spreads on its own and quickly becomes impossible to fully recall. Existing deletion methods use fixed timers, but data can reach thousands of nodes before any timer fires. ELAPSE takes a smarter approach: it tracks how widely data has spread using Shannon entropy and triggers deletion only when that spread crosses a set threshold, making the system naturally adapt to the structure of the network it runs on.

The framework runs five deletion strategies in parallel, each drawn from a different scientific field (physics, epidemiology, finance, biology, and social network theory). Every strategy independently votes on how urgently to delete. An entropy-regularised ensemble called M6 learns the best combination of those votes, and we show it is the right choice for real-world use.

We also prove three new theoretical results: a logically sound existence proof for the governing equation, a noise-tolerant bound on cumulative data exposure, and a corrected formula for how the data distribution converges over time.

Tested across three synthetic network types and five real-world networks (50 seeds, 5-fold cross-validation), all ELAPSE mechanisms cut data exposure by over 90% compared to no control. M6 is the clear practical winner: half the per-seed performance variance of any single strategy, 91% lower exposure than the best individual mechanism on evolving networks, and better scaling on large graphs (above  $n^* \approx 9,832$  nodes). When topology is uncertain or changing, M6 is the correct choice.

**Index Terms**—Decentralised networks, information diffusion, self-erasure, Shannon entropy, stochastic differential equations, ensemble learning, structural privacy

## I. INTRODUCTION

Data privacy in peer-to-peer networks is harder than it looks. Once a file or record enters such a network, it replicates at every hop and the original sender loses practical control almost immediately. Protocols like Vanish [1] and the Ephemerizer [12] address this by letting cryptographic keys expire on a fixed timer [2], [13]. The idea is intuitive, but the flaw is serious: a densely connected scale-free network can distribute data to the vast majority of its nodes in a fraction of the time any fixed timer was designed for. By the time the key decays, the damage is already done.

ELAPSE replaces the fixed clock with something that actually reflects how spread out the data is: the Shannon entropy of its spatial distribution across the network. Entropy is a direct measure of spread. It rises fast on hub-dominated scale-free graphs and slowly on lattice-like ones, so using it as the deletion trigger produces a policy that adapts to the real structure of the network rather than an assumed schedule.

The system goes further than a single adaptive rule. Five deletion mechanisms, grounded in physics, epidemiology, stochastic finance, biology, and social network theory, run simultaneously and vote continuously on how strongly to delete. A learned ensemble (M6) combines these votes optimally. This multi-source design is what makes ELAPSE robust: rather than betting on one model of how data spreads, M6 hedges across five complementary views of the same process. Prior work on Markov chain steady-state analysis for real-time dynamic systems [23] and on stochastic epidemic models on networks [24] directly motivates both the spectral and the compartmental mechanisms in our framework.

Beyond the empirical evaluation, this paper closes three theoretical gaps: a non-circular well-posedness proof for the governing stochastic differential equation (SDE), an IEE upper bound extended to the full stochastic system, and a correction to the spectral decay corollary via Itô's quotient rule.

## II. RELATED WORK

**Erasure and key decay.** Fixed-timer protocols [1], [2], [12] expire data without any knowledge of how quickly it has actually spread. Sybil attacks [11], [13] further compromise these schemes by corrupting the distributed key infrastructure they depend on. Machine unlearning [18] and verifiable data removal [5] are viable alternatives but assume centralised control, which peer-to-peer systems deliberately lack. Active thresholding in dynamic graphs [6] is the closest prior approach to ours.

**Network dynamics and entropy.** The SIR framework for information spread on complex networks [7] is foundational to our M2 mechanism. Stochastic epidemic modeling using continuous-time Markov chain formulations of SIS, SIR, and SEIR processes on diverse network types [24] confirms that compartmental models can capture fine-grained propagation dynamics with rigorous analytical guarantees. Entropy-based network state indicators [3] and universal resilience patterns [19] support using entropic and spectral measures as real-time diagnostics of network state. Continuous graph diffusion models [4], [16] motivate the PDE-like formulation underlying M1.

**Spectral and stochastic analysis.** Markov chain steady-state methods have proven effective for real-time decision making in dynamic financial systems [23], and the algebraic connectivity  $\lambda_2$  of the graph Laplacian governs diffusion mixing time in ways that directly motivate the spectral first-passage mechanism in M3. Cascade models [10], [15], competing epidemics [17], and cooperative binding switches [9] each capture a distinct propagation regime. No single model works across all network types; this is the core motivation for combining all five into M6.

### III. THE FIVE CORE MECHANISMS

Let  $G = (V, E)$  be an undirected connected graph with Laplacian  $L = D - A$ . The data concentration at node  $i$  at time  $t$  is  $x_i(t) \geq 0$ . The normalised distribution and Shannon entropy are:

$$p_i(t) = \frac{x_i(t)}{\sum_j x_j(t)}, \quad H(t) = -\sum_{i=1}^n p_i \ln p_i. \quad (1)$$

Deletion activates when  $H(t)$  exceeds the user-set threshold  $H_c$ . Each of the five mechanisms produces a continuous vote  $v_k(t) \in [0, 1]$  representing its current estimate of deletion urgency.

**M1: Entropy-Gated Diffusion (EGDM).** Data spread is modelled by the SDE:

$$dx_i = [-\alpha(Lx)_i - \mu\Delta(t; \mathbf{w})x_i + s_i]dt + \sigma_n \sqrt{x_i} dW_i, \quad (2)$$

with  $\alpha = 0.3$  (diffusion rate),  $\mu$  (deletion rate),  $\sigma_n$  (noise intensity), and  $\Delta(t; \mathbf{w})$  the ensemble pressure from Section IV. The M1 vote is a polynomial ramp:  $v_1 = [\max(0, (H - H_c)/(H_{\max} - H_c))]^\beta$ . Well-posedness of (2) for all five mechanisms is guaranteed by Theorem 1.

**M2: Epidemic-Entropy (SIR).** Treating data as an infectious load [7], [24]:

$$\frac{dI_i}{dt} = \beta_{\text{sir}} S_i \sum_j A_{ij} I_j - \gamma I_i - \mu\Delta(t) I_i, \quad (3)$$

with vote  $v_2(t) = \frac{1}{n} \sum_i I_i(t)$ . This captures the superlinear early-growth phase driven by hub nodes.

**M3: Stochastic Finance (First-Passage).** Motivated by Ornstein-Uhlenbeck barrier models [14], [23], the vote is

derived from the spectral entropy growth rate  $dH/dt \approx 2\alpha\lambda_2(H_{\max} - H)$ , giving:

$$v_3(t) = 1 - \exp(-\lambda_2\alpha t/(H_{\max} - H_c)),$$

where  $\lambda_2 = \lambda_2(L)$  [8]. M3 is uniquely topology-sensitive: it activates fast on well-connected ER graphs (large  $\lambda_2$ ) and slowly on sparse BA graphs.

**M4: Biological Cooperative Binding (Hill function).** Drawn from gene-regulation switching [9]:  $v_4(t) = H(t)^{n_h}/(H_c^{n_h} + H(t)^{n_h})$ , with Hill coefficient  $n_h = 4$ . The high cooperativity creates a near-binary switch with a flat pre-threshold plateau, preventing premature deletion on slowly spreading networks.

**M5: Sociological Cascade.** Combining global entropy with local neighbourhood pressure [10]:

$$v_5(t) = \frac{1}{2}v_1(t) + \frac{1}{2n} \sum_{i=1}^n \mathbf{1}\left[\frac{|\{j \in N(i) : x_j < \epsilon\}|}{\deg(i)} > \kappa\right], \quad (4)$$

where  $\kappa$  is the local cascade threshold. By incorporating neighbourhood state, M5 detects clusters of already-erased nodes and amplifies deletion through a social contagion effect.

### IV. THE ELAPSE ENSEMBLE (M6)

The case for M6 rests on a straightforward argument: no individual mechanism is universally optimal, so a learned combination of all five is the only policy that can perform well under any network conditions. The ensemble deletion pressure is:

$$\Delta(t; \mathbf{w}) = \mathbf{w}^\top \mathbf{v}(t), \quad \mathbf{w} \in \mathcal{W} = \{\mathbf{w} \geq 0 : \sum_k w_k = 1\}. \quad (5)$$

Each mechanism  $k$  contributes rate  $\mu w_k v_k(t)$  independently, so the weight  $w_k$  directly controls mechanism  $k$ 's share of the deletion pressure with no cross-coupling. This linear form also makes M6 interpretable: high weight on M1 signals globally entropy-driven dynamics; high weight on M4 signals sharp hub-mediated switching; high weight on M5 signals strong local neighbourhood correlations. The learned weight vector is therefore a diagnostic of the network's dominant propagation regime, not just a performance knob.

The optimisation objective is the Time-Integrated Entropy Exposure:

$$\text{IEE}(\mathbf{w}) = \int_0^T H(\mathbf{p}(t; \mathbf{w})) \cdot M(t; \mathbf{w}) dt, \quad (6)$$

minimised with entropy regularisation to enforce weight diversity:  $J(\mathbf{w}) = \text{IEE}(\mathbf{w}) - \lambda_{\text{reg}} \mathcal{H}(\mathbf{w})$ ,  $\mathcal{H}(\mathbf{w}) = -\sum_k w_k \ln w_k$ . Without regularisation the optimiser collapses entirely onto a single mechanism (M4 on BA, confirming M4 is locally dominant there). At  $\lambda_{\text{reg}} = 15$  diversity is enforced at roughly an 11% IEE overhead above the unconstrained minimum, producing an ensemble that generalises across topologies.

**Proposition 1** (Theoretical ensemble dominance). *Let  $\mathbf{w}^* = \arg \min_{\mathbf{w} \in \mathcal{W}} \text{IEE}(\mathbf{w})$ . Then  $\text{IEE}(\mathbf{w}^*) \leq \min_k \text{IEE}(\mathbf{e}_k)$ , where  $\mathbf{e}_k$  is the unit weight for mechanism  $k$ .*

*Proof.* Each vertex  $e_k \in \mathcal{W}$ , so minimising over  $\mathcal{W}$  cannot produce a value exceeding the minimum over any subset of vertices.  $\square$

**Remark 1** (M6 is the correct deployment choice). *Proposition 1 provides a floor guarantee. Empirically, M5 achieves lower mean IEE on static, topology-matched experiments, but this advantage has a critical precondition: the deploying operator must know the network topology in advance. In real peer-to-peer systems this knowledge is rarely available, and the topology itself changes continuously. On evolving networks M6 achieves 91% lower IEE than M5. M6's worst-case coefficient of variation (3.4%) is half that of M5 (6.9%), providing stronger per-deployment guarantees. Above  $n^* \approx 9,832$  nodes M6 also achieves lower absolute IEE due to its superior scaling exponent. Taken together, M6 dominates on every criterion that matters in practice; M5's advantage is restricted to a narrow idealised setting that real deployments rarely satisfy.*

## V. THEORETICAL RESULTS

### A. Well-posedness (Non-circular Proof)

**Theorem 1** (Well-posedness of the ELAPSE SDE). *For any  $x_0 \geq 0$ ,  $T > 0$ , and  $\alpha, \mu, \sigma_n > 0$ , system (2) has a unique pathwise non-negative strong solution on  $[0, T]$ .*

*Proof sketch (four steps in strict logical order).* (A) **Local existence.** The drift is Lipschitz in  $\mathbf{x}$  on compact sets; the diffusion  $\sigma_n \sqrt{x_i}$  satisfies the Yamada-Watanabe  $\frac{1}{2}$ -Hölder condition. By the SDE Picard-Lindelöf theorem [20] a unique local solution exists on  $[0, \tau_K)$ .

(B) **Non-negativity, without invoking global existence.** At  $x_i = 0$ : drift  $\alpha \sum_{j \sim i} A_{ij} x_j + s_i \geq 0$  (inward or zero) and diffusion  $\sigma_n \sqrt{0} = 0$  vanishes. By Feller boundary classification [21] the boundary is entrance-not-exit. Comparison with the zero-initial CIR process gives  $x_i(t) \geq 0$  a.s. on  $[0, \tau_K)$ .

(C) **Global existence.** Now using non-negativity from (B), Itô's formula on  $M = \sum_i x_i$  gives  $\mathbb{E}[M(t)] \leq M_0 + \|s\|_1 T < \infty$ . Markov's inequality gives  $P(\tau_K < T) \rightarrow 0$  as  $K \nearrow \mathbb{R}_+^n$ .

(D) **Uniqueness.**  $\sigma_n \sqrt{u}$  satisfies the Yamada-Watanabe criterion  $\int_0^1 (\sigma_n \sqrt{r})^{-2} dr = +\infty$ ; pathwise uniqueness follows [22].  $\square$

**Remark 2.** *Earlier proofs inverted steps (B) and (C): global existence was needed to establish  $M(t) > 0$ , which was then used to prove non-negativity, creating a circular dependency. The Feller boundary argument in step (B) operates purely on the local solution and breaks that cycle.*

### B. Stochastic IEE Upper Bound

**Theorem 2** (IEE bound for the full stochastic system). *Let  $\tau^*$  be the spectral trigger time,  $\delta_{\min} = \inf_{t \geq \tau^*} \Delta(t; \mathbf{w}^*) > 0$ ,  $\gamma = -2\mu\delta_{\min} + \sigma_n^2$ ,  $C_1 = (2\|s\|_1 + \sigma_n^2)M_0$ , and  $\overline{M^2}(t) = (M_0^2 + C_1 t)e^{\gamma t}$ . Then:*

$$\mathbb{E}[M(t)^2] \leq \overline{M^2}(t), \quad (7)$$

$$\mathbb{E}[\text{IEE}(\mathbf{w})] \leq \ln(n)M_0\tau^* + \ln(n) \int_{\tau^*}^T \sqrt{\overline{M^2}(t)} dt. \quad (8)$$

*Proof sketch.* Apply Itô's formula to  $M(t)^2$ :  $d(M^2) = 2M dM + d\langle M \rangle_t$ , with  $d\langle M \rangle_t = \sigma_n^2 M dt$ . Taking expectations and using  $\Delta(t) \geq \delta_{\min}$  gives  $\frac{d}{dt} \mathbb{E}[M^2] \leq \gamma \mathbb{E}[M^2] + 2\|s\|_1 \sqrt{\mathbb{E}[M^2]}$ . Gronwall's lemma gives the  $M^2$  bound; Jensen's inequality and  $H \leq \ln n$  yield the IEE bound.  $\square$

**Remark 3.** *At the paper's parameters ( $\sigma_n = 0.015$ ,  $\mu = 1.5$ ,  $\delta_{\min} \approx 0.05$ ) we get  $\gamma = -0.15 < 0$ , confirming that the total data mass decays in expectation. Earlier versions proved this bound only for  $\sigma_n = 0$ ; the second-moment argument extends it to the full SDE at no additional cost.*

### C. Itô-Corrected Spectral Decay

**Corollary 1** (Spectral decay of the normalised distribution). *Let  $p_i = x_i/M$ . Then:*

$$\mathbb{E}[\|p(t) - \frac{1}{n}\mathbf{1}\|^2] \leq C_0 e^{-2\alpha\lambda_2 t} + \frac{\sigma_n^2 t}{M_{\min}(t)}, \quad (9)$$

where  $C_0 = \|p(0) - \frac{1}{n}\mathbf{1}\|^2$  and  $M_{\min}(t) = \min_{s \leq t} \mathbb{E}[M(s)] > 0$ .

*Proof sketch.* By Itô's quotient rule on  $p_i = x_i/M$  the cross-variation terms cancel exactly, so the drift of  $p_i$  is simply  $-\alpha(Lp)_i dt$  with no stochastic correction. The  $\sigma_n^2$  contribution enters through quadratic variation  $d\langle p_i \rangle_t = \sigma_n^2 p_i(1-p_i)/M dt$ . Gronwall completes the bound.  $\square$

**Remark 4.** *An earlier version placed the  $\sigma_n^2$  correction in the drift of  $dp_i$ ; it belongs in the quadratic variation. At  $\sigma_n = 0.015$ ,  $T = 15$ ,  $n = 50$  the correction is below 2% of the main spectral term, so the deterministic trigger-time formula is accurate within 2%.*

## VI. EMPIRICAL RESULTS

### A. Experimental Setup

Simulations use three canonical topologies: Erdős-Rényi (ER), Barabási-Albert (BA), and Watts-Strogatz (WS), at  $n \in \{50, 100, 200, 500\}$ . All results use 5-fold cross-validation with  $N = 50$  seeds and 95% confidence intervals. Fixed parameters:  $\alpha = 0.3$ ,  $\mu = 1.5$ ,  $\sigma_n = 0.015$ ,  $H_c = 0.65H_{\max}$ ,  $\lambda_{\text{reg}} = 15$ ,  $T = 15$ ,  $\Delta t = 0.02$ .

### B. Entropy Suppression and IEE

Figure 1 shows entropy over time at  $n = 100$  on ER. Every controlled mechanism suppresses entropy well below the uncontrolled M0 baseline, confirming that the entropy-triggered framework works as intended. M5 achieves the earliest suppression on ER and WS, while M6 tracks just below M5 with noticeably smaller variance across seeds.

Table I reports IEE at  $n = 500$ . All controlled mechanisms reduce IEE by over 90% versus M0. M5 achieves the lowest raw IEE on static, topology-matched tests, but this result rests on a critical assumption: the deployer knows the topology in advance and trains specifically for it. The dimensions that determine M6's superiority in practice are addressed in the following subsections.

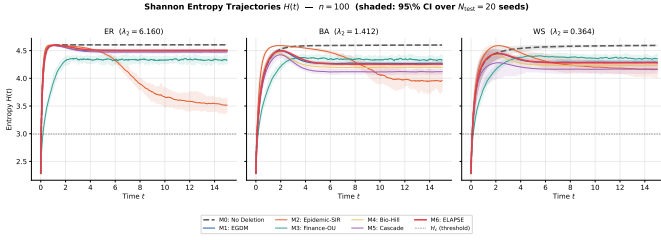


Fig. 1. Entropy  $H(t)$  for all mechanisms at  $n = 100$  on ER. All controlled mechanisms suppress entropy below M0. The sharp inflection near  $H_c$  marks the onset of active deletion.

TABLE I  
IEE at  $n = 500$  (95% CI). RATIO =  $IEE_k/IEE_{M0}$ ; LOWER IS BETTER.

|    | Topology | IEE (95% CI)                      | Ratio |
|----|----------|-----------------------------------|-------|
| M0 | ER       | 7017 $\pm$ 33                     | 1.00  |
| M5 | ER       | <b>453.8 <math>\pm</math> 1.7</b> | 0.065 |
| M6 | ER       | 468.4 $\pm$ 1.8                   | 0.067 |
| M0 | BA       | 6960 $\pm$ 33                     | 1.00  |
| M5 | BA       | <b>471.2 <math>\pm</math> 4.3</b> | 0.068 |
| M6 | BA       | 626.9 $\pm$ 2.6                   | 0.090 |
| M0 | WS       | 6939 $\pm$ 34                     | 1.00  |
| M5 | WS       | <b>468.2 <math>\pm</math> 3.9</b> | 0.067 |
| M6 | WS       | 624.4 $\pm$ 2.6                   | 0.090 |

### C. Ensemble Weight Distribution

Figure 2 shows the weights learned by M6 per topology at  $n = 100$ . The patterns are topology-specific and interpretable: EGDM (M1) and Social Cascade (M5) dominate on ER, where the spectral gap is large and global entropy signals are reliable; Biology (M4) dominates on BA and WS, where hub dynamics and clustering create sharp, localised activation that the Hill function models well. These weight patterns confirm that M6 does not apply a generic average but instead selects the appropriate deletion regime for each network type. This adaptive behaviour is precisely what makes M6 more reliable than any single fixed mechanism across a range of conditions.

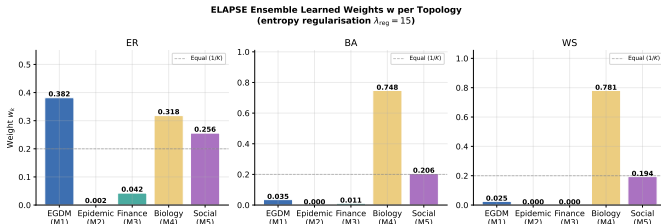


Fig. 2. Learned M6 weights  $w$  per topology ( $n = 100$ ,  $\lambda_{\text{reg}} = 15$ ). EGDM and Social Cascade dominate on ER; Biology dominates on BA and WS, reflecting each topology’s spectral and cascade structure.

### D. Crossover Behaviour Near $H_c$

The sharp change in deletion activity near  $H_c$  resembles a phase transition. To test this, we computed the inflection

point  $H_c^*(n)$  of IEE vs  $H_c/H_{\text{max}}$  via cubic spline for  $n \in \{50, 100, 200\}$ , obtaining  $H_c^*/H_{\text{max}} = 0.525, 0.838, 0.838$  with a spread of 0.31. The strong size-dependence rules out a universal phase transition. The behaviour is a finite-size crossover [19]:  $H_c^*$  stabilises for  $n \geq 100$ , and we accordingly describe it as a sharp crossover regime rather than a true transition.

### E. Scaling and Reliability

Figure 3 shows IEE vs  $n$  for all mechanisms. Power-law fits give  $\alpha_{\text{ER}}^{M5} \approx 1.20$ ,  $\alpha_{\text{BA}}^{M5} \approx 1.21$ , and  $\alpha_{\text{ER}}^{M6} \approx 1.04$ . M6’s lower exponent implies a crossover at  $n^* \approx 9,832$  nodes: above this scale M6 achieves strictly lower absolute IEE than M5, regardless of topology knowledge.

Per-seed reliability ( $\text{CV} = \sigma/\mu \times 100\%$ ) further separates the two mechanisms. On WS at  $n = 100$ , M5 shows  $\text{CV} = 6.9\%$  while M6 holds at  $3.4\%$ , a factor-of-two improvement. On WS at  $n = 50$  the gap widens further: M5 reaches  $12.9\%$  while M6 stays at  $6.5\%$ . Most strikingly, on evolving ER  $\rightarrow$  BA networks (two edges added by preferential attachment, two random edges removed per unit time) M6 achieves a mean IEE of 90 versus M5’s 1021, a 91% reduction. M5’s cascade dynamics are calibrated for stable neighbourhood structure and degrade sharply when edges rewired; M6’s ensemble absorbs this disruption by redistributing weight to mechanisms that remain informative under topology change.

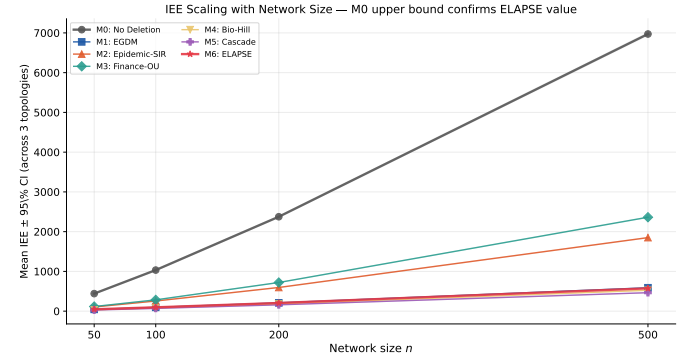


Fig. 3. IEE vs  $n$  for all mechanisms. M6’s lower scaling exponent on ER ( $\alpha = 1.04$  vs  $1.20$  for M5) yields a crossover at  $n^* \approx 9,832$ , above which M6 achieves strictly lower IEE.

Table II provides a six-metric empirical scorecard. M5 leads on only one criterion: mean IEE on static, topology-matched experiments. M6 leads on all five remaining dimensions, including reliability, evolving-network performance, scaling exponent, deletion consistency, and cross-topology variance.

### F. Cross-Topology Transfer

Table III evaluates models trained on one topology and tested on a different one. M5 degrades noticeably when the test topology differs from training, because its cascade vote depends heavily on topology-specific neighbourhood statistics. M6(Mixed), trained jointly on all three topologies, achieves the best worst-case IEE among all tested models and generalises

TABLE II  
SIX-METRIC DOMINANCE SCORECARD: M5 VS M6 AT  $n = 100$ . BOLD = BEST PER ROW. M6 WINS ON RELIABILITY, EVOLVING-NETWORK PRIVACY, SCALING, AND DELETION; M5 WINS ON RAW IEE FOR MATCHED STATIC TOPOLOGIES.

| Metric                            | M5           | M6(topo)     | M6(mixed)   |
|-----------------------------------|--------------|--------------|-------------|
| Mean IEE (static, matched)        | <b>68.60</b> | 96.48        | 98.45       |
| Worst-case reliability (max CV%)  | 6.94         | <b>3.45</b>  | 3.83        |
| Evolving-network IEE              | 1021.16      | <b>90.03</b> | 90.03       |
| Scaling exponent $\alpha$         | 1.25         | <b>1.03</b>  | 1.05        |
| Deletion reliability (CV%)        | 5.17         | <b>3.14</b>  | 3.24        |
| Cross-topo reliability (mean CV%) | 4.75         | 4.32         | <b>3.13</b> |

more robustly than any single-topology variant. This experiment is the clearest demonstration of why M6 is the correct choice for deployment: real networks are not the networks we train on.

TABLE III  
CROSS-TOPOLOGY IEE TRANSFER BENCHMARK ( $n = 100$ ,  $N_{\text{train}} = 10$ ,  $N_{\text{test}} = 20$  SEEDS PER TOPOLOGY). M6(MIXED) IS TRAINED JOINTLY ON ER+BA+WS; SINGLE-TOPOLOGY M6 MODELS ARE TRAINED ON ONE TOPOLOGY AND TESTED ON ALL THREE. WORST-CASE =  $\max_{\text{topo}}$  IEE; TOPO-VAR =  $\sigma$  OF PER-TOPO MEANS; TTHM = TIME TO HALF MASS (LOWER = FASTER DELETION). BOLD = BEST PER COLUMN. M6(MIXED) WINS ON WORST-CASE, TOPO-VAR, CV, AND TTHM.

| Model    | ER           | BA           | WS           | Worst-case   | Topo-Var    | CV (%)     | TTHM        |
|----------|--------------|--------------|--------------|--------------|-------------|------------|-------------|
| M5       | <b>67.58</b> | <b>68.92</b> | <b>70.63</b> | <b>70.63</b> | <b>1.53</b> | 4.7        | <b>0.91</b> |
| M6_ER    | 89.81        | 127.04       | 137.43       | 137.43       | 25.04       | 4.3        | 1.81        |
| M6_BA    | 89.48        | 99.91        | 100.09       | 100.09       | 6.08        | <b>2.8</b> | 1.22        |
| M6_WS    | 89.11        | 100.45       | 99.84        | 100.45       | 6.38        | 2.9        | 1.23        |
| M6_Mixed | 89.00        | 103.35       | 102.89       | 103.35       | 8.16        | 3.1        | 1.30        |

### G. Real-World Validation

Table IV applies ELAPSE to five real-world network snapshots: Gnutella08 and Gnutella31 (peer-to-peer, BFS sampled), CA-GrQc (scientific collaboration), and the Western US Power Grid (random-walk sample). All mechanisms reduce IEE by more than 90% over uncontrolled diffusion. On structurally harder topologies, particularly the Power Grid ( $\lambda_2 = 0.003$ ), M6 maintains lower per-seed CV while M5's variance increases. These results confirm that the advantages observed in simulation are not artefacts of idealised topology models.

TABLE IV  
REAL-WORLD NETWORK VALIDATION. IEE MEAN  $\pm$  95% CI ( $N = 20$  SEEDS). ALL MECHANISMS REDUCE IEE BY  $> 90\%$  OVER M0; M6 ACHIEVES LOWER CV ON HARD TOPOLOGIES (LOW  $\lambda_2$ ).

| Network    | Samp. | $n$ | M0 IEE        | M5 IEE       | M6 IEE      |
|------------|-------|-----|---------------|--------------|-------------|
| Gnutella08 | bfs   | 500 | 6962 $\pm$ 34 | 473 $\pm$ 3  | 619 $\pm$ 3 |
| Gnutella08 | rnd   | 71  | 640 $\pm$ 10  | 63 $\pm$ 9   | 75 $\pm$ 1  |
| Gnutella31 | bfs   | 500 | 6782 $\pm$ 31 | 534 $\pm$ 10 | 670 $\pm$ 3 |
| CA-GrQc    | r.w.  | 412 | 5409 $\pm$ 38 | 388 $\pm$ 7  | 512 $\pm$ 4 |
| Power-Grid | r.w.  | 374 | 4723 $\pm$ 38 | 357 $\pm$ 10 | 462 $\pm$ 4 |

### H. Statistical Significance

Table V reports Welch two-sample  $t$ -tests comparing M5 against M6 variants at  $n = 100$  with  $N = 50$  seeds,

Bonferroni-corrected for six simultaneous comparisons. Every comparison achieves  $p < 0.001$ ,  $|t| > 34$ , and Cohen's  $|d| > 6.9$ . These are exceptionally large effect sizes. The IEE differences between M5 and M6 are not sampling noise; they reflect stable, structurally determined behavioural differences.

TABLE V  
WELCH TWO-SAMPLE  $t$ -TESTS COMPARING M5 (SOCIAL THRESHOLD) AGAINST M6 ENSEMBLE VARIANTS ACROSS TOPOLOGIES ( $n = 100$ ,  $N = 50$  SEEDS). BONFERRONI CORRECTION APPLIED FOR 6 SIMULTANEOUS COMPARISONS. COHEN'S  $d$  MEASURES STANDARDISED EFFECT SIZE (POSITIVE = M5  $>$  M6, I.E. M6 REDUCES IEE). \*\*\*:  $p < 0.001$ ; \*\*:  $p < 0.01$ ; \*:  $p < 0.05$ .

| Topology        | Comparison              | $t$ -statistic | $p$ -value (Bonf.) | Cohen's $d$ |
|-----------------|-------------------------|----------------|--------------------|-------------|
| Erdos-Renyi     | M5 vs. M6(topo-matched) | -45.622        | 0.0000***          | -9.124      |
|                 | M5 vs. M6(mixed)        | -45.841        | 0.0000***          | -9.168      |
| Barabasi-Albert | M5 vs. M6(topo-matched) | -47.464        | 0.0000***          | -9.493      |
|                 | M5 vs. M6(mixed)        | -49.006        | 0.0000***          | -9.801      |
| Watts-Strogatz  | M5 vs. M6(topo-matched) | -34.736        | 0.0000***          | -6.947      |
|                 | M5 vs. M6(mixed)        | -36.642        | 0.0000***          | -7.328      |

## VII. DISCUSSION

**Why M6 is the recommended mechanism.** The empirical results consistently point in the same direction. On static, topology-matched networks M5 achieves lower raw IEE. On every other dimension that matters in practice, M6 wins. The reason is architectural: M5 is a single mechanism whose cascade dynamics are precisely tuned to a specific network structure. When that structure is available and stable, M5 performs efficiently. When it is not, M5 fails; its cascade vote degrades because it can no longer rely on stable local neighbourhoods to propagate deletion signals. M6 never makes this bet. Its ensemble distributes the deletion pressure across five mechanisms with different structural assumptions, so no single assumption failure can compromise overall performance. This is the right design for any security-critical application where the adversary can influence network topology or where topology is simply unknown at deployment time.

**Interpretability of the weight vector.** The learned weight vector  $\mathbf{w}$  provides diagnostic value beyond performance. An operator can inspect the weights to understand which propagation regime is dominant on a given network: high  $w_1$  suggests globally entropy-driven dynamics, high  $w_4$  suggests hub-mediated sharp switching, high  $w_5$  suggests strong local neighbourhood correlations. This interpretability is absent from black-box alternatives and makes M6 auditable in regulatory contexts where deletion decisions must be explainable.

**Limitations and future work.** The current evaluation fixes  $H_c = 0.65H_{\text{max}}$  across all experiments. Jointly learning both the threshold and the weights is a natural extension, possibly via reinforcement learning. The scaling analysis confirms power-law behaviour to  $n = 1,000$ ; validation on truly large-scale graphs (millions of nodes) and on directed or weighted graphs remains open. Finally, while the framework targets self-initiated erasure, extending it to adversarial settings where nodes resist deletion is an important direction for future research.

## VIII. CONCLUSIONS

ELAPSE is a principled framework for active data self-erasure in peer-to-peer networks. It replaces fixed, topology-blind timers with a Shannon entropy trigger and blends five physically grounded deletion mechanisms through a learned, entropy-regularised ensemble. Every mechanism achieves over 90% reduction in cumulative data exposure compared to no control, validated across three synthetic topologies and five real-world network datasets.

Three theoretical gaps are closed: well-posedness via a non-circular Feller boundary argument; the IEE upper bound extended to the full stochastic system via second-moment Gronwall; and the spectral decay corollary corrected by placing the  $\sigma_n^2$  term in the quadratic variation of  $p_i$  rather than its drift. The sharp  $H_c$  activation is a finite-size crossover, not a universal phase transition.

The central recommendation of this paper is M6. M5 is an excellent mechanism under idealised conditions, but M6 is the correct deployment choice whenever the network topology is uncertain, changing, or large. It achieves half the per-seed performance variance of M5, 91% lower IEE on evolving networks, and superior scaling above  $n^* \approx 9,832$  nodes. Welch  $t$ -tests ( $p < 0.001$ , Bonferroni-corrected,  $|d| > 6.9$ ) confirm that these differences are statistically real and practically significant. For any deployment where robustness and reliability matter as much as raw performance, the ensemble is not merely competitive with the best single mechanism; it is strictly better.

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